

Extended Vehicle Speed Control HLD

Document ID: HLD-VSC-LARGE-001 **Version:** 1.0 **Status:** Synthetic Demonstration Document

Purpose: This synthetic document is intended only for scalability and retrieval demonstration of the AUTOSAR HLD Document Analysis Assistant. It is not an OEM production HLD.

1. Executive Overview

This High-Level Design describes a synthetic vehicle-speed control architecture used to demonstrate long-document ingestion and semantic retrieval. It covers sensing, speed processing, display, diagnostics, warning, logging, gateway adaptation and health supervision.

The architecture is organized around explicit software components, typed interfaces, provided/required ports, signals, and component dependencies. The document intentionally includes repeated engineering context around unique architecture facts so that semantic retrieval can be exercised on a longer document.

2. Architecture Summary

Component	Responsibility	Key Interface	Primary Dependency
WheelSpeedSensor	Acquires wheel-speed information from the sensing layer.	WheelSpeedInterface	No upstream software component; provides wheel-speed data.
VehicleSpeedController	Processes wheel-speed information and derives vehicle-speed information.	VehicleSpeedInterface	Requires WheelSpeedSensor.
InstrumentCluster	Displays vehicle-speed information to the driver.	DisplayStatusInterface	Requires VehicleSpeedController.
DiagnosticsManager	Monitors vehicle-speed processing status and exposes diagnostic status.	DiagnosticsStatusInterface	Requires VehicleSpeedController.
SpeedWarningManager	Evaluates configurable speed thresholds and requests warnings.	SpeedWarningInterface	Requires VehicleSpeedController.
DataLogger	Records selected speed and diagnostic information for engineering analysis.	EngineeringDataInterface	Requires VehicleSpeedController and DiagnosticsManager.
GatewayAdapter	Adapts selected vehicle-speed information for external networks.	GatewaySpeedInterface	Requires VehicleSpeedController.
HealthSupervisor	Aggregates health status for the speed-processing chain.	SystemHealthInterface	Requires DiagnosticsManager and WheelSpeedSensor.

3. Component Design - WheelSpeedSensor

Responsibility. Acquires wheel-speed information from the sensing layer. The component participates in the vehicle speed control architecture and exchanges data through explicitly defined AUTOSAR-style interfaces.

Interface behavior. The principal interface for this component is **WheelSpeedInterface**. Its architecture relationship is documented so that provider/consumer consistency can be checked automatically.

Dependency. WheelSpeedSensor has the following primary dependency context: No upstream software component; provides wheel-speed data.. Dependency information is used by impact analysis, functional-flow construction and architecture graph generation.

Port Name	Type	Interface	Description
WheelSpeedSensorOut	P-Port	WheelSpeedInterface	Provides WheelSpeedInterface data where applicable.
WheelSpeedSensorIn	R-Port	SensorHealthInterface	Consumes SensorHealthInterface input where applicable.

3.1 Engineering Notes

WheelSpeedSensor is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 1: verify that WheelSpeedInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement

without sending the full document to the language model.

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Review focus 2: verify that WheelSpeedInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

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4. Component Design - VehicleSpeedController

Responsibility. Processes wheel-speed information and derives vehicle-speed information. The component participates in the vehicle speed control architecture and exchanges data through explicitly defined AUTOSAR-style interfaces.

Interface behavior. The principal interface for this component is **VehicleSpeedInterface**. Its architecture relationship is documented so that provider/consumer consistency can be checked automatically.

Dependency. VehicleSpeedController has the following primary dependency context: Requires WheelSpeedSensor.. Dependency information is used by impact analysis, functional-flow construction and architecture graph generation.

Port Name	Type	Interface	Description
VehicleSpeedControllerOut	P-Port	VehicleSpeedInterface	Provides VehicleSpeedInterface data where applicable.
VehicleSpeedControllerIn	R-Port	WheelSpeedInterface	Consumes WheelSpeedInterface input where applicable.

4.1 Engineering Notes

VehicleSpeedController is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

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5. Component Design - InstrumentCluster

Responsibility. Displays vehicle-speed information to the driver. The component participates in the vehicle speed control architecture and exchanges data through explicitly defined AUTOSAR-style interfaces.

Interface behavior. The principal interface for this component is **DisplayStatusInterface**. Its architecture relationship is documented so that provider/consumer consistency can be checked automatically.

Dependency. InstrumentCluster has the following primary dependency context: Requires VehicleSpeedController.. Dependency information is used by impact analysis, functional-flow construction and architecture graph generation.

Port Name	Type	Interface	Description
InstrumentClusterOut	P-Port	DisplayStatusInterface	Provides DisplayStatusInterface data where applicable.
InstrumentClusterIn	R-Port	VehicleSpeedInterface	Consumes VehicleSpeedInterface input where applicable.

5.1 Engineering Notes

InstrumentCluster is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 1: verify that DisplayStatusInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

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Review focus 4: verify that DisplayStatusInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement

without sending the full document to the language model.

6. Component Design - DiagnosticsManager

Responsibility. Monitors vehicle-speed processing status and exposes diagnostic state. The component participates in the vehicle speed control architecture and exchanges data through explicitly defined AUTOSAR-style interfaces.

Interface behavior. The principal interface for this component is **DiagnosticStatusInterface**. Its architecture relationship is documented so that provider/consumer consistency can be checked automatically.

Dependency. DiagnosticsManager has the following primary dependency context: Requires VehicleSpeedController.. Dependency information is used by impact analysis, functional-flow construction and architecture graph generation.

Port Name	Type	Interface	Description
DiagnosticsManagerOut	P-Port	DiagnosticStatusInterface	Provides DiagnosticStatusInterface data where applicable.
DiagnosticsManagerIn	R-Port	VehicleSpeedInterface	Consumes VehicleSpeedInterface input where applicable.

6.1 Engineering Notes

DiagnosticsManager is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

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Review focus 4: verify that DiagnosticStatusInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

7. Component Design - SpeedWarningManager

Responsibility. Evaluates configurable speed thresholds and requests warning indication. The component participates in the vehicle speed control architecture and exchanges data through explicitly defined AUTOSAR-style interfaces.

Interface behavior. The principal interface for this component is **SpeedWarningInterface**. Its architecture relationship is documented so that provider/consumer consistency can be checked automatically.

Dependency. SpeedWarningManager has the following primary dependency context: Requires VehicleSpeedController.. Dependency information is used by impact analysis, functional-flow construction and architecture graph generation.

Port Name	Type	Interface	Description
SpeedWarningManagerOut	P-Port	SpeedWarningInterface	Provides SpeedWarningInterface data where applicable.
SpeedWarningManagerIn	R-Port	VehicleSpeedInterface	Consumes VehicleSpeedInterface input where applicable.

7.1 Engineering Notes

SpeedWarningManager is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 1: verify that SpeedWarningInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

7.2 Engineering Notes

SpeedWarningManager is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 2: verify that SpeedWarningInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

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Review focus 4: verify that SpeedWarningInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This

context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

8. Component Design - DataLogger

Responsibility. Records selected speed and diagnostic information for engineering analysis. The component participates in the vehicle speed control architecture and exchanges data through explicitly defined AUTOSAR-style interfaces.

Interface behavior. The principal interface for this component is **LoggingInterface**. Its architecture relationship is documented so that provider/consumer consistency can be checked automatically.

Dependency. DataLogger has the following primary dependency context: Requires VehicleSpeedController and DiagnosticsManager.. Dependency information is used by impact analysis, functional-flow construction and architecture graph generation.

Port Name	Type	Interface	Description
DataLoggerOut	P-Port	LoggingInterface	Provides LoggingInterface data where applicable.
DataLoggerIn	R-Port	DiagnosticStatusInterface	Consumes DiagnosticStatusInterface input where applicable.

8.1 Engineering Notes

DataLogger is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 1: verify that LoggingInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

8.2 Engineering Notes

DataLogger is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

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Review focus 4: verify that LoggingInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

9. Component Design - GatewayAdapter

Responsibility. Adapts selected vehicle-speed information for external network consumers. The component participates in the vehicle speed control architecture and exchanges data through explicitly defined AUTOSAR-style interfaces.

Interface behavior. The principal interface for this component is **GatewaySpeedInterface**. Its architecture relationship is documented so that provider/consumer consistency can be checked automatically.

Dependency. GatewayAdapter has the following primary dependency context: Requires VehicleSpeedController.. Dependency information is used by impact analysis, functional-flow construction and architecture graph generation.

Port Name	Type	Interface	Description
GatewayAdapterOut	P-Port	GatewaySpeedInterface	Provides GatewaySpeedInterface data where applicable.
GatewayAdapterIn	R-Port	VehicleSpeedInterface	Consumes VehicleSpeedInterface input where applicable.

9.1 Engineering Notes

GatewayAdapter is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 1: verify that GatewaySpeedInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

9.2 Engineering Notes

GatewayAdapter is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 2: verify that GatewaySpeedInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

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Review focus 4: verify that GatewaySpeedInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This

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10. Component Design - HealthSupervisor

Responsibility. Aggregates health status for the speed-processing chain. The component participates in the vehicle speed control architecture and exchanges data through explicitly defined AUTOSAR-style interfaces.

Interface behavior. The principal interface for this component is **SystemHealthInterface**. Its architecture relationship is documented so that provider/consumer consistency can be checked automatically.

Dependency. HealthSupervisor has the following primary dependency context: Requires DiagnosticsManager and WheelSpeedSensor.. Dependency information is used by impact analysis, functional-flow construction and architecture graph generation.

Port Name	Type	Interface	Description
HealthSupervisorOut	P-Port	SystemHealthInterface	Provides SystemHealthInterface data where applicable.
HealthSupervisorIn	R-Port	DiagnosticStatusInterface	Consumes DiagnosticStatusInterface input where applicable.

10.1 Engineering Notes

HealthSupervisor is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 1: verify that SystemHealthInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

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Review focus 2: verify that SystemHealthInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

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Review focus 4: verify that SystemHealthInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

11. Signal Definitions

Signal	Interface	Unit	Description
WheelSpeed	WheelSpeedInterface	km/h	Measured wheel-speed value.
VehicleSpeed	VehicleSpeedInterface	km/h	Calculated vehicle-speed value.
DiagnosticStatus	DiagnosticStatusInterface	enum	Diagnostic state of the speed-processing chain.
SpeedWarning	SpeedWarningInterface	boolean	Speed-threshold warning state.
SystemHealth	SystemHealthInterface	enum	Aggregated health indication.

12. Functional Flows

Flow 1: WheelSpeedSensor -> VehicleSpeedController -> InstrumentCluster

The flow is expressed at component level for architecture review. It can be used to reason about upstream providers, downstream consumers, change impact and test-planning scope.

Flow 2: VehicleSpeedController -> DiagnosticsManager -> HealthSupervisor

The flow is expressed at component level for architecture review. It can be used to reason about upstream providers, downstream consumers, change impact and test-planning scope.

Flow 3: VehicleSpeedController -> SpeedWarningManager

The flow is expressed at component level for architecture review. It can be used to reason about upstream providers, downstream consumers, change impact and test-planning scope.

Flow 4: VehicleSpeedController -> GatewayAdapter

The flow is expressed at component level for architecture review. It can be used to reason about upstream providers, downstream consumers, change impact and test-planning scope.

Flow 5: VehicleSpeedController -> DataLogger

The flow is expressed at component level for architecture review. It can be used to reason about upstream providers, downstream consumers, change impact and test-planning scope.

13. Interface Governance

Interface Governance consideration 1. The vehicle speed architecture shall preserve explicit ownership of interfaces and traceability between component responsibilities, ports, signals and dependencies. Reviewers should verify that architecture changes remain visible in revision comparison and that document-specific questions can be supported by retrieved evidence. This synthetic paragraph represents the type of contextual engineering prose that can surround structured facts in a longer HLD.

Interface Governance consideration 2. The vehicle speed architecture shall preserve explicit ownership of interfaces and traceability between component responsibilities, ports, signals and dependencies. Reviewers should verify that architecture changes remain visible in revision comparison and that document-specific questions can be supported by retrieved evidence. This synthetic paragraph represents the type of contextual engineering prose that can surround structured facts in a longer HLD.

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Interface Governance consideration 4. The vehicle speed architecture shall preserve explicit ownership of interfaces and traceability between component responsibilities, ports, signals and dependencies. Reviewers should verify that architecture changes remain visible in revision comparison and that document-specific questions can be supported by retrieved evidence. This synthetic paragraph represents the type of contextual engineering prose that can surround structured facts in a longer HLD.

Interface Governance consideration 5. The vehicle speed architecture shall preserve explicit ownership of interfaces and traceability between component responsibilities, ports, signals and dependencies. Reviewers should verify that

architecture changes remain visible in revision comparison and that document-specific questions can be supported by retrieved evidence. This synthetic paragraph represents the type of contextual engineering prose that can surround structured facts in a longer HLD.

Interface Governance consideration 6. The vehicle speed architecture shall preserve explicit ownership of interfaces and traceability between component responsibilities, ports, signals and dependencies. Reviewers should verify that architecture changes remain visible in revision comparison and that document-specific questions can be supported by retrieved evidence. This synthetic paragraph represents the type of contextual engineering prose that can surround structured facts in a longer HLD.

14. Diagnostics and Observability

Diagnostics and Observability consideration 1. The vehicle speed architecture shall preserve explicit ownership of interfaces and traceability between component responsibilities, ports, signals and dependencies. Reviewers should verify that architecture changes remain visible in revision comparison and that document-specific questions can be supported by retrieved evidence. This synthetic paragraph represents the type of contextual engineering prose that can surround structured facts in a longer HLD.

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15. Timing and Data Freshness

Timing and Data Freshness consideration 1. The vehicle speed architecture shall preserve explicit ownership of interfaces and traceability between component responsibilities, ports, signals and dependencies. Reviewers should verify that architecture changes remain visible in revision comparison and that document-specific questions can be supported by retrieved evidence. This synthetic paragraph represents the type of contextual engineering prose that can surround structured facts in a longer HLD.

Timing and Data Freshness consideration 2. The vehicle speed architecture shall preserve explicit ownership of interfaces and traceability between component responsibilities, ports, signals and dependencies. Reviewers should verify that architecture changes remain visible in revision comparison and that document-specific questions can be supported by retrieved evidence. This synthetic paragraph represents the type of contextual engineering prose that can surround structured facts in a longer HLD.

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16. Fault Handling

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17. Change Impact Analysis

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Change Impact Analysis consideration 2. The vehicle speed architecture shall preserve explicit ownership of interfaces and traceability between component responsibilities, ports, signals and dependencies. Reviewers should verify that architecture changes remain visible in revision comparison and that document-specific questions can be supported by retrieved evidence. This synthetic paragraph represents the type of contextual engineering prose that can surround structured facts in a longer HLD.

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18. Verification Considerations

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19. Deployment Assumptions

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20. Traceability Notes

Traceability Notes consideration 1. The vehicle speed architecture shall preserve explicit ownership of interfaces and traceability between component responsibilities, ports, signals and dependencies. Reviewers should verify that architecture changes remain visible in revision comparison and that document-specific questions can be supported by retrieved evidence. This synthetic paragraph represents the type of contextual engineering prose that can surround structured facts in a longer HLD.

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Appendix A - Architecture Inventory

WheelSpeedSensor: responsibility=Acquires wheel-speed information from the sensing layer.; interface=WheelSpeedInterface; dependency=No upstream software component; provides wheel-speed data..

VehicleSpeedController: responsibility=Processes wheel-speed information and derives vehicle-speed information.; interface=VehicleSpeedInterface; dependency=Requires WheelSpeedSensor..

InstrumentCluster: responsibility=Displays vehicle-speed information to the driver.; interface=DisplayStatusInterface; dependency=Requires VehicleSpeedController..

DiagnosticsManager: responsibility=Monitors vehicle-speed processing status and exposes diagnostic state.; interface=DiagnosticStatusInterface; dependency=Requires VehicleSpeedController..

SpeedWarningManager: responsibility=Evaluates configurable speed thresholds and requests warning indication.; interface=SpeedWarningInterface; dependency=Requires VehicleSpeedController..

DataLogger: responsibility=Records selected speed and diagnostic information for engineering analysis.; interface=LoggingInterface; dependency=Requires VehicleSpeedController and DiagnosticsManager..

GatewayAdapter: responsibility=Adapts selected vehicle-speed information for external network consumers.; interface=GatewaySpeedInterface; dependency=Requires VehicleSpeedController..

HealthSupervisor: responsibility=Aggregates health status for the speed-processing chain.; interface=SystemHealthInterface; dependency=Requires DiagnosticsManager and WheelSpeedSensor..

Appendix B - Scalability Test Notes

This document is deliberately longer than the controlled accuracy benchmark. It should be used to demonstrate ingestion, chunk creation, vector indexing and retrieval behavior. It should not be merged into the existing precision/recall/F1 benchmark unless a separate page-level/entity-level ground truth and evaluation protocol are defined.